Description of the invention*:*

The present invention is of a handheld portable and manually operated water filtration device designed for the removal of arsenic, heavy metals and turbidity from contaminated groundwater or surface water including flood water. It integrates the principles of the Arsiron Nilogon method with a novel piston-driven filtration mechanism contained within a single, compact bottle as shown in Figure 1.

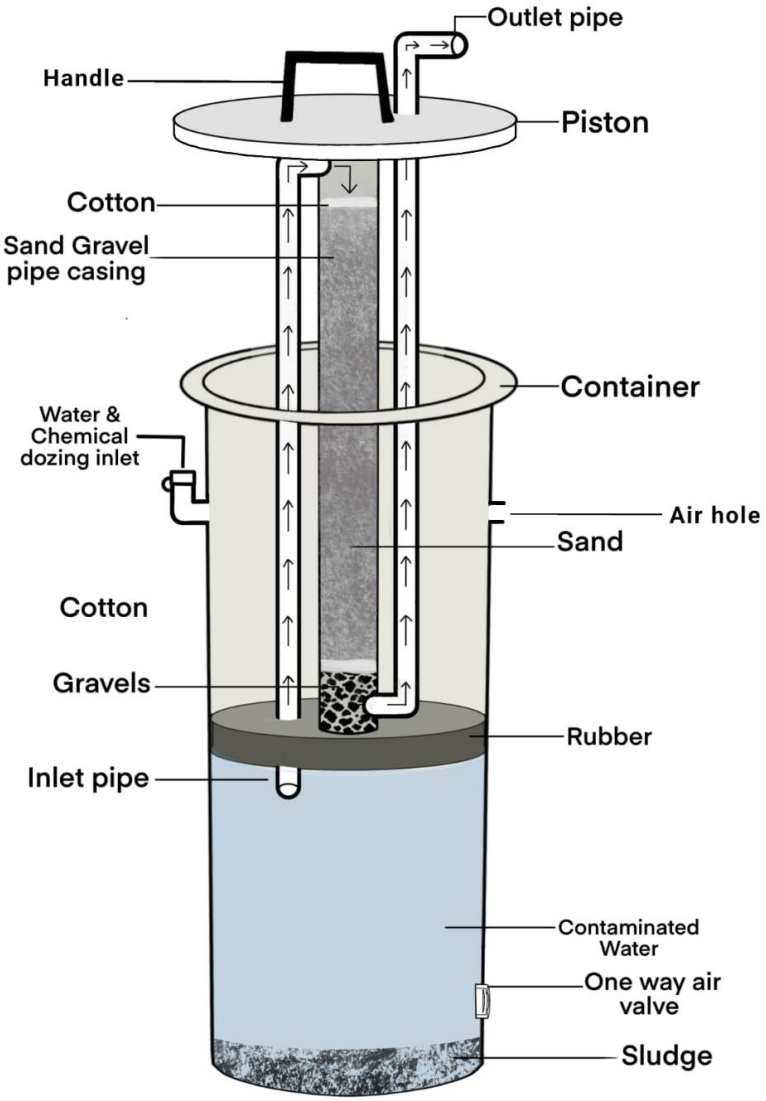


Figure 1: Schematic diagram of piston water bottle model.

The system comprises a bottle-like cylindrical transparent plastic container that holds ½ litre to 1 litre of the contaminated water. Treatment begins with the addition of sodium bicarbonate ($NaHCO_3$) to adjust the pH, potassium permanganate ($KMnO_4$) to oxidize arsenite {As(III)} to arsenate {As(V)}, and ferrous iron to insoluble ferric oxides. The added ferric chloride ($FeCl_3$)

to form ferric oxide-hydroxide flocs. These flocs adsorb and co-precipitate arsenic, co-precipitate other heavy metals and removes turbidity through coagulation- sedimentation. The resulting sludge settles at the bottom of the container.

Once sedimentation is complete, a piston assembly is manually pushed into the container. The piston is fitted with:

(1) A vertical sand–gravel filter pipe (1/2 inch to 1 inch diameter),

(2) An inlet pipe connected to the top of the filter pipe (4-8mm inner 4-diameter),

(3) An outlet pipe connected to the bottom of the filter pipe (8mm inner diameter).

These three components the filter pipe, inlet pipe, and outlet pipe are fixed to the piston and move vertically as a single unit.

When the user presses the piston downward, mechanical pressure is created. The inlet pipe, which gets submerged in the water, draws in chemically treated but unfiltered water. This water flows through the inlet and enters the top of the sand-gravel filter pipe, where it passes through graded filtration media (fine sand to coarse gravels). As the water travels downward through the filter, particulates and flocs are trapped. The resulting clean water is pushed out through the outlet pipe, which is connected to the base of the filter pipe and discharges outside the bottle for direct collection or consumption. This configuration ensures that:

- No re-contamination occurs, as the inlet sits above the settled sludge.
- Filtration is inline and pressure-assisted, enabling faster flow than gravity-fed models.
- No electricity or external pump is needed, making it ideal for on-the-go users.

A dedicated water and chemical dosing inlet is introduced at the upper left side of the container. This inlet is equipped with a permanently attached cap. It allows the user to fill water and dose chemicals without needing to remove the piston assembly. The piston rests just above the level of this inlet when at the top position, enabling seamless filling before starting the filtration. An air hole is provided on the upper right side of the container, functioning as an air release during the filling process. This helps eliminate trapped air when water is added through the new inlet. The hole remains open by default but is positioned in such a way that, once the piston is pushed downward, the air hole remains above the piston, avoiding interference during filtration. A one-way air valve is installed at the lower right side of the container. This valve allows only air to enter the container when the piston is pulled back up after use. It prevents vacuum formation inside the container, making it easier to reset the piston for reuse without requiring high manual effort.

The invention is optimized for single-user operation and is especially suitable for defence personnel, students, and travellers operating in arsenic or heavy metal-affected or remote areas.

The device provides a compact, efficient, and low-cost method of producing safe drinking water from chemically treated sources.

An advantage of this filter over straw filters is that the water here can be collected in a glass or cup before drinking which allows a visual check of removal of some purity like absence of sediments, suspended solid, colour or turbidity (which can be indicator of presence of contaminants like iron oxides) in addition to removing arsenic and heavy metal ions. Its unique integration of chemical treatment, piston-driven flow, and sand-gravel filtration in a mobile, reusable format distinguishes it from conventional filters and establishes its novelty in the field of water purification.

Technical details of the filter bottle:

Container specification-

- Capacity- 500ml plastic bottle
- Diameter – 7 cm
- Height – 20.6 cm

Plastic Piston specification-

- Height – 22.5 cm
- Diameter – 7 cm
- Surface area – 3.848×10^{-3} m^2 or 38.48 cm^2

Filter pipe specification-

- Length – 19.05 cm
- Diameter – ¾ inch or 1.905 cm

Inlet and outlet pipe specification-

- Length – 23 cm
- Diameter – 8mm

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Letter of Recommendation

For Mr. Aarav A. Jain

This is my pleasure to write about Mr. Aarav A. Jain who did internship with me during Summer 2025.

Mr. Aarav emailed me with a proposal to work with me a couple of months before meeting at Tezpur University during the summer. I was initially hesitant to allow him to work in my laboratory as he was too young to do research. However, I had to change my mind after he had met me in the university. Throughout my career, I have noticed, a student's dedication is more important than his talent for the kind of research I have been doing. I had been impressed by his enthusiasm and interest in reaching out to rural people through safe water which is my main research area.

During his internship Mr. Aarav contributed towards designing in developing a handheld bottle filter, developed by my research group, that removed arsenic other heavy metals and turbidity in addition to regulating the pH of water using the already patented Arsiron Nilogon method. This has enabled him to include his name among the inventors of a patent entitled, "Handheld portable filter for removal of arsenic, heavy metal ions and turbidity from drinking water", filed in the Indian Patent Office, with application number of 202531130896, dated 23 December 2025.

As I saw a proficiency in AI in Mr. Aarav, I advised him to make an automated dosing unit for a small community, say 200L-1000L Arsiron Nilogon filter. He was able to make it and successfully demonstrate it in a demo Arsiron Nilogon unit at the University campus. Mr. Aarav continues to be in touch with me after the internship and utilises some of his time in working towards helping rural people in getting arsenic-free safe water.

Finally, I can say that I have not only seen a dedication in Mr. Aarav to his work, but also has an inquisitive and talented mind which he proactively applies in whatever he does. We all need to nurture such a talented and dedicated boy.

09/4/2026

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